AWT COMPETITION GUIDELINES 2025

OUR MISSION

TO PROMOTE
REWARDING MANUFACTURING
CAREERS FOR THE FUTURE OF OUR COMMUNITIES

@AWT RoboBots
www.awtrobobots.com

Revision 2- 10/13/25

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1. INTRODUCTION

1.1. AWT ROBOBOTS & JUNIOR ROBOBOTS

1.1.1. The Alliance for Working Together (AWT) was started in 2002 as a small, informal group of manufacturers who met to discuss topics of interest and best practices among local manufacturers. Finding skilled workers was quickly identified as one of the top challenges for our manufacturers. Promoting rewarding careers in manufacturing became the core mission of the organization.

1.1.2. In 2012, the AWT Foundation became a 501 (c)3 non-profit. Since then, the AWT Foundation has partnered with 500+ manufacturing companies, community organizations, and school members, all engaged in promoting rewarding manufacturing careers!

1.1.3. The student teams are encouraged to build partnerships with local manufacturing companies and post-secondary institutions during the bot-building process. This provides the students with direct application of classroom concepts to real-world manufacturing situations.

1.2. YEARLY COMPETITIONS

1.2.1. The dates, times, and locations of AWT RoboBots and Junior RoboBots competitions are determined by the organization each year and are posted on its website. Unless publicized otherwise, the competition is held on the last Saturday in April at Lakeland Community College.

1.3. CONTACTING ROBOBOTS & JUNIORBOTS

1.3.1. For questions, comments, requests, and clarifications regarding the competition rules, technical regulations, and procedures in any AWT documents, please contact AWT.

1.3.2. Program Coordinator: Monica Hammil

1.3.3. Address: 8980 Tyler Blvd, Mentor, OH 44060

1.3.4. Email: monica_hammil@thinkmfg.com

1.3.5. Website: https://thinkmfg.com/robobots/

2. GENERAL INFORMATION

2.1. GENERAL INFORMATION

2.1.1. RoboBots

2.1.1.1. 15lb bots compete against each other in a head-to-head match.

2.1.1.2. Matches will last 3 minutes or until one bot taps out or is knocked out.

2.1.1.3. The competition will be a double-elimination style tournament.

2.1.1.4. This competition also includes Documentation and Interviews

2.1.2. JuniorBots

2.1.2.1. 3lb bots compete against each other in a head-to-head match.

2.1.2.2. Matches will last 90 seconds until one bot taps out or is knocked out.

2.1.2.3. The competition will be a double-elimination style tournament.

2.1.2.4. This competition does not participate in Documentation and Interviews.

2.2. CONDUCT

2.2.1. All teams are expected to be responsible, safe, follow AWT rules, and be respectful. Fighting, unruly behavior, and disrespectful behavior will result in immediate expulsion from the competition.

2.2.2. Any cheating before or during the competition will result in immediate disqualification from the competition for the team. Depending on the severity of the misconduct, the judges with AWT will determine final outcomes.

2.2.3. It is assumed that the students do the majority of the work on the bot with adult supervision when necessary.

2.2.4. Non-students, including advisors, coaches, and parents, may not work on the bot unless they get approval from AWT staff or safety officials, or if intervening to prevent an imminent safety violation.

2.2.5. Inadvertently touching the bot while giving instructions, or holding the bot while a student works, does not count as working on the bot.

2.3. CLASSES & CLUBS

2.3.1. The intent behind the competition is for RoboBots (15lb) to be for High School students only and for JuniorBots (3lb) to be for Middle School students only.

2.3.2. High School is defined as students approximately 14-18 years of age or in grades 9-12.

2.3.3. Middle School is defined as students approximately 11-13 years of age or in grades 6-8.

2.4. TEAMS

2.4.1. A team is defined by its name and its affiliated school or youth organization.

2.4.2. A team includes at least two students and at least one adult Coach. Teams should also have an Industry Advisor.

2.4.3. There is no limit to the number of students allowed on a team; however, if the number exceeds 8, please contact AWT staff at least one month prior to the Competition to ensure adequate space in the pit area. AWT cannot guarantee space for team members beyond 8 in the pit.

2.4.4. A school can only have a total of four teams per weight class, and students cannot participate on multiple teams.

2.5. The Coach or Teacher is responsible for the students and must be at least 21 years old and affiliated with the school or organization.

2.6. An Industry Advisor is a professional who assists the team and offers them advice. They may be from a local company, post-secondary institution, or parent. There is no limit to the number of teams an Industry Advisor helps or the number of Industry Advisors a team can have.

2.7. Each Student must have contributed to the design and construction of the bot, and may be asked questions about the bot by an AWT representative during a competition.

2.7.1. **A Student may only be a member of one team. If a Student is enrolled at a school with a RoboBots or JuniorBots team, they should first try to compete with that team before forming a new or separate team.**

2.7.2. To form a team based in a qualified youth organization, contact the AWT for eligibility requirements. Note: the AWT reserves the right to request verification of current student enrollment.

2.8. BOT ELIGIBILITY

2.8.1. AWT highly encourages teams to design and build new bots for each competition season.

2.8.2. Any team wishing to re-enter a past bot will be required to submit a description of design improvements or changes that have been made in the previous year to be eligible to compete.

2.8.3. A bot that wins the First Place Award twice is automatically retired.

2.8.4. Changes should include making new components of some nature for the bot. Scope of changes required for re-entry of an existing bot includes the redesign and fabrication of at least one (Drive train components, Chassis design, Weapon design, or Weapon drive system).

2.8.5. Teams are required to show their bot design changes from year one to year two in their Documentation if using a similar design and improving on the same bot from one year to another. Teams cannot use the same bot for more than two years in a row.

2.8.6. JUNIOR ROBOTS ELIGIBILITY

2.8.6.1. Students can not participate on multiple teams (within the same school or different organizations).

2.8.6.2. Schools are limited to four teams per school.

2.9. RESOLVING PROBLEMS

2.9.1. If any issues need to be resolved, they should be brought to the attention of the AWT Coordinator. If the issue is a Protest, the rules outlined in Section 8.0 of this document will be followed.

2.9.2. For all other issues, the Program Coordinator will bring the issue(s) to the AWT Board President and AWT Executive Director to discuss a resolution. This will be completed as soon as possible. If the AWT Board President and AWT Executive Director are unavailable, the Program Coordinator will decide the outcome. The decision will be final.

2.10. INSURANCE

2.10.1. All events will have basic accident insurance coverage. Teams are advised to have their own coverage when not at an event.

2.11. PHOTOGRAPHS AND VIDEOTAPING

2.11.1. AWT allows individuals to photograph and videotape the competitions. Individuals are encouraged to display the AWT logo in said photos and videos.

2.11.2. Teams will sign a waiver stating their consent to videotaping and photography while at the competition.

2.12. COMPETITION AWARDS

2.12.1. The following RoboBots winners will be recognized at the competition: First Place, Second Place, Third Place, Best Documentation, Best Engineered, and Best Interview.

2.12.2. The following JuniorBots winners will be recognized at the competition: First Place, Second Place, and Third Place.

3. REGISTRATION REQUIREMENTS

3.1. REGISTRATION AGREEMENT

3.1.1. The AWT Registration Agreement is used to document the organization, team name, bot name, and the names of the participants. This document must be turned into AWT before competing.

3.1.2. All team members, including advisors and volunteers, must also submit a waiver and photo agreement before the competition.

3.2. BOT TRANSPORT

3.2.1. Once a team has checked in and its bot has passed safety inspection, the bot must remain taped down on its stand so that its wheels do not touch the table in the pit area until its first match.

3.2.2. Bots must remain in the pit area when not competing.

3.2.3. The only exception is to consult first with a safety official and have the bot tested.

3.3. DOCUMENTATION

3.3.1. Engineering Documentation is used to explain the process that a team went through to build their bot; this documentation process also serves as a learning exercise. The documentation includes: schematics, drawings, photos, timelines, journals, etc., that show the process of building your bot.

3.3.2. There is a mandatory documentation training at the beginning of November and a mandatory documentation check-in in late January/early February. Documentation must be submitted by the deadline set by AWT. These dates are provided to teams during the Kick-Off.

3.3.3. A team that fails to provide documentation for its bot is ineligible to compete.

3.4. TEAM INTERVIEWS

3.4.1. As part of the documentation review process, documentation judges will interview team members. Every team member should be prepared to

discuss the entire bot build process, including planning, design, construction, and strategy.

3.4.2. A team that fails to participate in interviews is ineligible to compete.

4. SAFETY RULES

4.1. SAFETY GLASSES

4.1.1. Safety glasses must be worn at all times when in the pit area, as well as in the arena area when your bot is competing.

4.1.2. Standard prescription glasses do not count as safety glasses. Certified prescription safety glasses with side shields or safety glasses designed to fit over prescription lenses are acceptable. Goggles are also acceptable.

4.1.3. Standard safety glasses should not be worn over prescription lenses. This rule also applies to coaches and technical advisors.

4.1.4. AWT will have safety glasses, overshield safety glasses, and goggles available. Safety is the responsibility of everyone. Non-compliance will result in a safety offense.

4.2. SMOKING PROHIBITED

4.2.1. Smoking is prohibited on Lakeland Community College's property, including outdoor spaces. This includes e-cigarettes and vapes.

4.3. ALCOHOL AND DRUGS

4.3.1. The use of alcohol, recreational, and illegal drugs is prohibited anywhere at the Competition.

4.4. PIT AREA RESTRICTIONS

4.4.1. Only team members are allowed in the pit area. Team members will be required to wear the ID provided by the event. This could simply be a wristband or a name badge.

4.4.2. There may be restrictions on the number of people allowed in the pit area, depending on the venue. Teams with over 8 students should check with the Event Organizer before a competition to find out the pit restrictions. Aisles must be kept clear of stationary objects. If needed, teams may use the area under the table for storage.

4.5. CLOTHING

4.5.1. Everyone in the pit area is required to wear appropriate clothing, and this includes closed-toed shoes and long pants.

4.5.2. Students, teachers, or advisors without appropriate clothing will be escorted from the pit area.

4.6. ADULT SUPERVISION

4.6.1. RoboBots:

4.6.1.1. When any team member is working on a bot, a supervising adult needs to be present. An adult can supervise no more than 2 teams at a time. High School students, even those over 18 years of age, are not eligible to provide adult supervision. The coach is responsible for their team members at all times.

4.6.2. JuniorBots:

4.6.2.1. When any team member is working on a bot, a supervising adult needs to be present. An adult can supervise no more than 1 team at a time. High School students, even those over 18 years of age, are not eligible to provide adult supervision. The coach is responsible for their team members at all times.

4.6.3. WELDING/GRINDING

4.6.3.1. Welding and grinding are not allowed in the pit area. They will be allowed in a designated area. Welding and grinding may be performed in the welding area, and must be supervised by a coach or other supervising adult.

5. MATCHES

5.1. MATCH LENGTH

5.1.1. Please reference section 2.1.1 for RoboBots and 2.1.2 JuniorBots.

5.2. TOURNAMENT PLACEMENT

5.2.1. The placement of your bot in the tournament brackets will be random. Minor adjustments may be made to ensure teams from the same school do not compete against each other in the first round.

5.3. JUDGING

5.3.1. Matches are judged on three criteria: aggression, control, and damage. For more information on judging, please refer to the Match Rules & Procedures and Judging Criteria documents.

5.4. GRUDGE MATCH

5.4.1. If the Event Organizer feels there is time, they may announce the opportunity for a “Grudge Match.” Teams will have the opportunity to sign up to fight a specific opponent.

5.4.2. Both teams are required to agree to the match, and both bots must have been eliminated from the competition to be eligible. The match time is standard for each weight class. Judges will decide the winner; this is purely for fun, and it will not affect any tournament results.

5.5. RUMBLE

5.5.1. If the Event Organizer feels there is time, they may announce the opportunity for a “Rumble.” Teams will then be offered the opportunity to sign up and fight against an unspecified number of opponents. All the teams entering their bots must have already been eliminated from the competition.

5.5.2. The “Rumble” will last for three minutes, and the bot with the most functionality at the end of the “Rumble” event will be declared the winner. Tap-outs will not be permitted. This is purely for fun, and this event will not affect any tournament results.

6. INSPECTION PROCEDURE

6.1. SAFETY INSPECTOR

6.1.1. Each Competition will have a Lead Safety Inspector. This person will be responsible for scheduling the inspections and keeping the inspection sheets for each bot. This person will be the main contact for any Safety inspection-related questions or comments.

6.2. INTERNAL/FUNCTIONAL INSPECTION

6.2.1. Each bot must pass both an “Internal and Functional Inspection” before competing. An “Internal Inspection” involves inspection of all of the components in a bot. A “Functional Inspection” involves a demonstration that the bot can safely be controlled and meets the fail-safe requirement listed in the Technical Regulations. A copy of the inspection form will be provided to each team at registration to use as a checklist.

6.3. OFFICIAL SCALE

6.3.1. AWT will have a specific scale designed as the official scale of the event. The team's Robot must be at class weight or under using this scale.

6.4. INSPECTION TIME SCHEDULE

6.4.1. Safety Inspection will take place during a specified time period before the start of the competition. This will be posted prior to the event.

6.5. AWT RIGHT TO INSPECT AND DISQUALIFY

6.5.1. AWT reserves the right to inspect or re-inspect your bot at any time during the competition. Any additions or changes to the bot must be re-inspected before competing, as well as after any significant damage during a match. If an AWT official feels a bot is unsafe for any reason and it cannot be made safe, it will be disqualified and not allowed to compete.

6.6. MATCH WEIGH-IN

6.6.1. All bots will be reweighed before each match.

6.7. NON-COMPLIANT

6.7.1. If, as a result of re-inspection, a bot does not comply with the Rules and Regulations, AWT officials will determine the outcome as follows:

6.7.1.1. If the non-compliance was not intentional and would not have affected the outcome of a previous match in which it fought, the previous match results will stand; however, the bot will need to be modified to make it compliant for future matches.

6.7.1.2. If the non-compliance was not intentional, but could have affected the outcome of a previous match in which it fought, the AWT officials can disqualify that bot and reverse the decision of the previous match if necessary. The bot will then need to be modified to make it compliant for future matches, if that loss did not eliminate them from the competition.

6.7.1.3. If the non-compliance was intentional, the bot will be disqualified, and any previous wins will be reversed. The non-compliant bot will

not be allowed to compete in any AWT competition for the entire year.

6.7.1.4. If the non-compliance was intentional and endangered any person(s) at the competition, the entire team will be asked to leave the competition. That team will not be allowed to compete in any AWT competition for the entire year.

6.8. BUILDING CAUTIONS

6.8.1. Combat robotic systems can be dangerous if not designed, constructed, and tested properly. Damage during matches can render the bot unsafe. It is ultimately the responsibility of the robotics teams to ensure the safety of their system design.

7. RULES ENFORCEMENT

7.1. RULES COMPLIANCE

7.1.1. In all matters of compliance with the Rules and any applicable civil or criminal laws, AWT and its officials reserve the right to penalize, or disqualify a bot, or to warn or expel any team or individual from the competition.

7.2. EXPELLED INDIVIDUAL

7.2.1. If an individual is expelled, they must leave the premises and return any and all IDs, pit passes, etc. This individual will not be permitted to compete in any future NRL event for the remainder of the year. If the individual expelled is the adult supervisor, the entire team must also leave the premises.

7.3. EXPELLED TEAM

7.3.1. If an entire team is expelled, they will be asked to leave the premises and return any and all IDs, pit passes, etc. They will also need to clear their pit table. They will not be allowed to compete in any NRL event for the remainder of the year.

7.4. CRIMINAL OFFENSES

7.4.1. If any person is in apparent violation of any Federal, State, or Local laws, that person will be turned over to the local authorities.

8. PROTESTS

8.1. ALLOWABLE PROTESTS

8.1.1. The following items may be protested:

8.1.1.1. The Safety Inspector’s decision to fail a bot during inspection

8.1.1.2. A match forfeit due to non-availability

8.1.1.3. Disqualification due to an inadvertent rules violation

8.2. UNALLOWABLE PROTESTS

8.2.1. The following items may not be protested:

8.2.1.1. The Judge’s decision on a match

8.2.1.2. Disqualification due to an intentional safety violation

8.2.1.3. Disqualification due to failure to obey an NRL official's instruction

8.3. PROTEST PROCEDURE

8.3.1. The procedure for protesting is as follows:

8.3.1.1. The protest must be lodged within 15 minutes of the time the event occurred.

8.3.1.2. The protest should be brought to the Program Coordinator.

8.3.1.3. All parties involved will be given a maximum of 5 minutes to present their side of the case.

8.3.1.4. After all parties have presented their side of the case, the Program Coordinator will make the final decision within one hour after the presentations.

8.3.1.5. The decision made by the Program Coordinator will be final.